Superseding
MIL-R-16920A
2 October 1952

MILITARY SPECIFICATION

RUBBER FACING, FOR RAMMER HEADS

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the facing for the rammer shell guards of 5-inch antiaircraft gun mounts.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

ZZ-H-499	Hose, Pneumatic, Wrapped
ZZ-T-381	Tires, Pneumatic, Vehicle and Portable Equipment

Military

MIL-P-116	Preservation, Methods of
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery

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STANDARDS

Federal

FED-STD-601	Rubber: Sampling and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking of Shipments

(When requesting any of the applicable documents, refer to both title and number. All request should be made via the cognizant Government inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code DCI), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Consolidated Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

American Trucking Association

National Motor Freight Classification Rules and Container Regulations

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N. W., Washington 6, D. C.)

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3. REQUIREMENTS

3.1 Material -

3.1.1 Rubber - Rubber shall be natural, reclaimed, or synthetic, of a grade suitable for use in standard automobile pneumatic tire threads (see Specification ZZ-T-381).

3.1.2 Backing - Backing, if used, shall be two-ply cotton woven duck, conforming to the material requirements of Specification ZZ-H-499.

3.2 Construction - The facing is to be molded to the form shown on Figure 1. A backing of two-ply cotton woven duck to reinforce the facing for attachment to the rammer head shall be optional. When furnished, it shall conform to the requirements of Specification ZZ-H-499, except that the plies need not be laid at 45° to one another. Both reinforced and unbacked facing shall be required to withstand the same physical tests.

* 3.3 Compression - Each sample facing must be able to withstand the compression test described in paragraph 4.5.1.

3.4 Impact - Each sample facing must be able to withstand, successfully, the impact test described in paragraph 4.5.2.

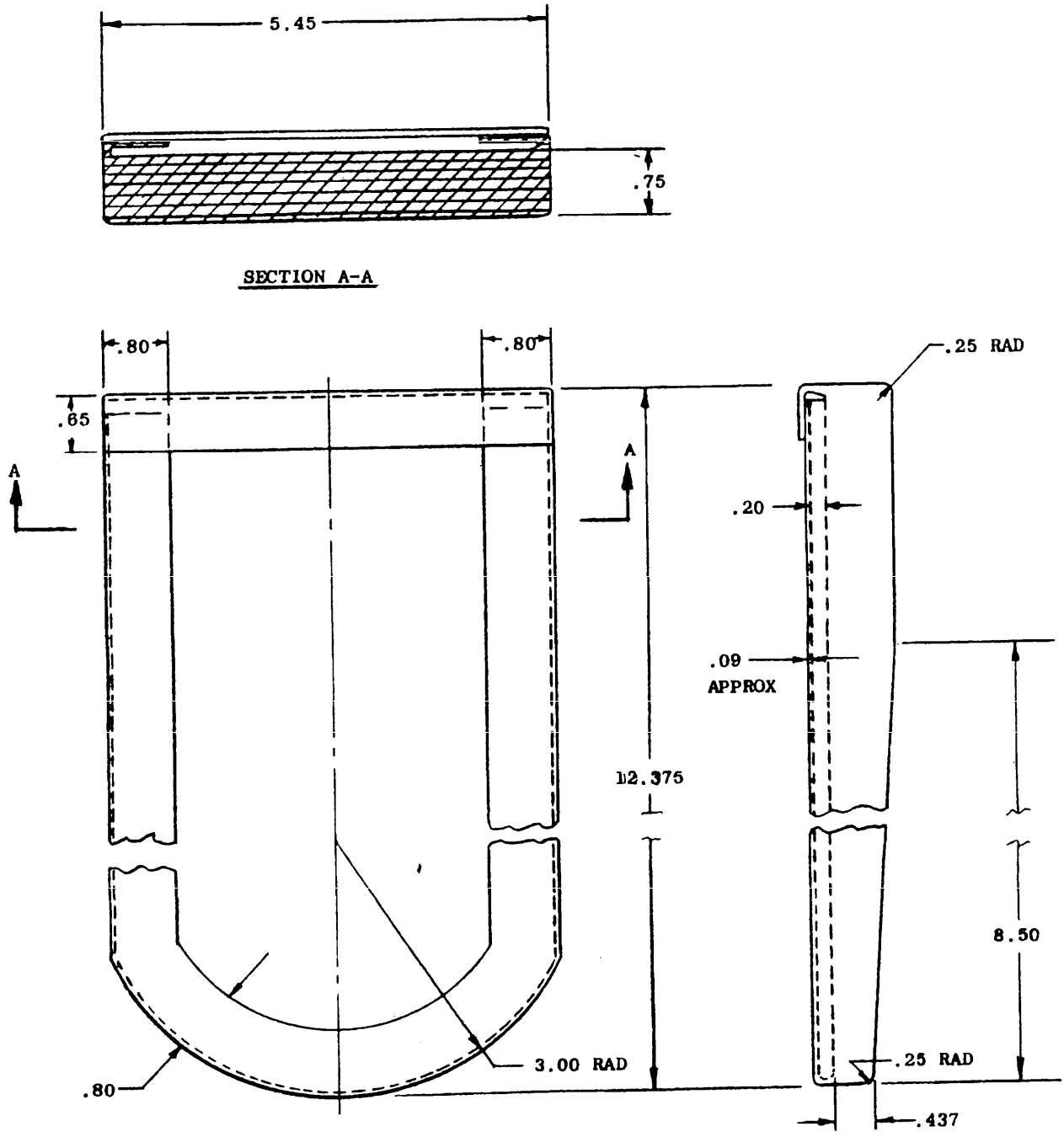
* 3.5 Hardness - The rubber shall have a Shore A Durometer hardness of 70 ± 5 points between 0°F and 180°F (-18° to 82°C) when tested as specified in paragraph 4.5.3.

* 3.6 Workmanship - Workmanship shall be of a quality to assure materials comply with the requirements of this specification.

* ## 4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

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DIMENSIONS IN INCHES.

FIGURE 1. JIG FOR FACING

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4.2 Classification of inspection -

4.2.1 Quality conformance inspection - Quality conformance tests for acceptance of individual lots shall consist of tests for all requirements specified in Section 3.

4.3 Sampling - Select a sample from each lot in accordance with MIL-STD-105, Inspection Level S-3, acceptance number zero.

4.4 Test conditions - Unless otherwise specified, tests shall be conducted at a temperature of 75° ± 5°F (24° ± 3°C) with a relative humidity of 50 ± 5 percent.

4.5 Test methods -

4.5.1 Compression test - The facing shall be tested in the thinnest section, under a load of 2,250 pounds applied by an indenter having a cross-section of 3 square inches, and a rounded edge on the indenting face to prevent cutting. There shall be no sign of permanent set or cutting after application of the load for 30 seconds.

4.5.2 Impact test - An impact of 350 foot-pounds shall be applied to the assembled facing 300 times, through a disc equivalent to a cartridge case base and at a rate not in excess of 2 per minute. This test, when performed on the area used in ramming, shall show no cutting, separation of fabric or failure in the fastening flaps sufficiently serious to render the facing unsuitable for further use. (This facing is assembled on a shell guard element that fits between the flaps and facing proper and is secured thereto by twenty-three No. 10 machine screws that act through the bearing strips and flaps into the tapped shell guard element.)

4.5.3 Hardness - Hardness shall be determined in accordance with Method 3021 of Federal Test Method Standard No. 601, using the Shore A Durometer.

4.6 Packaging, packing, and marking - Preparation for delivery shall be examined for conformance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Packaging shall be level A or C as specified in the contract or order, in accordance with the provisions outlined in Specification MIL-P-7936. Level A material shall be packaged in accordance with Specification MIL-P-116, method III.

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* 5.2 Packing - Packing shall be level A, B, or C as specified in the contract or order, in accordance with the procedures outlined in Specification MIL-P-7936.

5.3 Marking - In addition to any special marking required by the contract or order, marking shall be in accordance with Military Standard MIL-STD-129.

6. NOTES

* 6.1 Intended use - The rubber facing covered by this specification is intended to cushion the impact between the rammer head and the shell casing.

* 6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Construction of facing (see 3.2).

(c) Quantity required.

(d) Applicable levels of packaging and packing required (see Section 5).

* 6.3 Changes from previous issue - The outside margins of this document have been marked "*" to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-R-16920B(Weps), Rubber Facing, For Rammer Heads

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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